

Vaasu Sohee

Data Scientist

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EDUCATION

M.S. Data Science · Michigan State University · GPA: 3.92 / 4.0

Aug 2024 – May 2026

B.Tech Computer Science · NIIT · GPA: 3.6 / 4.0

Aug 2019 – May 2023

EXPERIENCE

Data Scientist Intern (Co-op) · Delta Dental Insurance

Jun 2025 – Present

- Architected a Provider Insights AI platform (Dataiku, Snowflake, Azure OpenAI, LLM Mesh) with governed metrics and join paths, enabling accurate NL analytics across 131 datasets.
- Built a GPT-5.5 mini + Dataiku LLM Mesh pipeline (OCR, few-shot prompting, Pydantic validation) extracting structured fields from thousands of handwritten/non-standard invoices and eliminating hundreds of manual hours.
- Engineered high-volume Oracle SQL + Dataiku pipelines for multimillion-row healthcare datasets, using SQL query optimization and partition-aware processing for actuarial forecasting and government program analytics.
- Led a self-supervised transformer embedding framework (PyTorch Lightning, contrastive pretraining, Optuna, MLflow) on non-CUDA Intel Arc GPUs, improving recall by 40%+ versus baselines across 3M+ healthcare claims.

Data Scientist · Ernst & Young

Jan 2023 – Jun 2024

- Directed Fortune 500 legacy on-prem-to-cloud migrations using AWS SageMaker, managed feature stores, and automated retraining, reducing retraining cycle time by 30% and improving inference accuracy by 15%.
- Re-engineered enterprise ETL pipelines (Python, SQL, Apache Airflow) processing 5M+ daily records, cutting latency from 4+ hours to under 30 minutes and enabling real-time insights for executive stakeholders.
- Designed A/B tests and uplift models on Fortune 100 datasets, identifying growth levers that delivered a 30% revenue KPI lift across client portfolios.
- Containerized and deployed scalable ML inference services using Docker and FastAPI across multi-region AWS environments, accelerating deployment velocity by 25% with zero-downtime rollouts.

Data Science Intern · Ernst & Young

Jun 2022 – Aug 2022

- Built and deployed cross-validated classification and regression models across 10+ client engagements, delivering a 20% improvement in submission rates and improving decision quality by 25% for Operations and Marketing teams.
- Applied uplift modeling and controlled experiments to provide advisory insights to client leadership, consulting on data-driven strategies for non-technical stakeholders.

Data Science Intern · NIIT

Jun 2021 – Dec 2021

- Developed a CNN-based deep learning model in PyTorch achieving 95%+ accuracy, deployed with real-time inference pipelines at sub-100ms latency and automated drift monitoring dashboards.

LATEST PROJECTS

AgentGuard - Security gateway for AI agents · [Link](#) · Independent Project

May 2026 – Aug 2026

- Built an AI agent tool runtime gateway (MCP, Next.js, TypeScript) enforcing 5 runtime controls: provenance checks, policy rules, secret stripping, human approval for risky actions, and tamperproof audit logging.
- Integrated OpenAI Agents SDK, LangGraph, Mastra, and PydanticAI agents with a shared backend via LiteLLM; stress tested against 4 realworld attacks: prompt injection, secret leaks, unauthorized rollbacks, and unauthorized access.
- Built a 3part policy intelligence layer: semantic search (Qdrant/LlamaIndex), audit history (PostgreSQL), and classifier tuning/tracing (DSPy, Arize Phoenix).

Demand & Capacity Forecasting Capstone · (Signed NDA) · Jackson National Life

Aug 2025 – Dec 2025

- Built a multi segment forecasting pipeline using Temporal Fusion Transformer and SARIMAX with exogenous regressors across 40+ contract segments, reducing RMSE by 22% and MAPE by 18% versus baseline.
- Translated model outputs into \$3.5M in annual risk reduction through optimized hedging strategies, crediting rate guarantees, and RILA pricing decisions in collaboration with business stakeholders.

[Portfolio](#) - More projects

SKILLS

Cloud & Data Infrastructure: AWS (S3, SageMaker, Athena, Bedrock), Snowflake, Docker, Kubernetes

Data Engineering & Pipelines: ETL/ELT pipeline design, Apache Airflow, Apache Spark, Dataiku

Programming: Python (Pandas, NumPy, scikit-learn), SQL (Oracle, PostgreSQL)

ML & MLOps: PyTorch, MLflow, FastAPI, CI/CD, LangChain, LangGraph, RAG, Prompt Engineering, Pydantic validation

LLM Evaluation & Observability: LLM-as-a-Judge, Ragas, LangSmith, TruLens, Azure OpenAI

Analytics & Modeling: Demand forecasting, A/B testing, Causal inference, Predictive analytics